
Model-Driven Engineering for Reliable REST APIs

Equipe : DiverSE

Site web de l'équipe : <http://www.irisa.fr/diverse/>

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Unité de recherche : IRISA - UMR 6074

Description du sujet de la thèse :

REST APIs have become the dominant mechanism for exposing web services across a wide range of domains, including cloud computing, e-commerce, healthcare, finance, and public administration. Despite their widespread adoption, the design and evolution of REST APIs remain complex and error-prone. In practice, inconsistencies frequently arise between business requirements, API implementations, and API specifications. For example, an API specification may describe a parameter as optional while the implementation requires it, or expose data structures that violate design rules. Such inconsistencies can lead to integration failures, vulnerabilities, unreliable behavior, maintenance difficulties, and poor API quality. The objective of this thesis is to **identify, prevent, and mitigate design, consistency, and compliance issues in REST APIs through specification-driven engineering approaches**. The thesis focuses on OpenAPI specifications as formal contracts between service providers and consumers and investigates how these specifications can be enriched with explicit constraints related to API quality, business rules, and regulatory policies.

Objectives of this doctorate

The objectives of this doctoral thesis are formulated through the following research questions:

- **RQ1:** How can design, consistency, and compliance issues in REST APIs be systematically identified at the specification level?
- **RQ2:** How can business rules, quality constraints and requirements be explicitly represented in REST API specifications to improve API reliability and prevent unintended behavior?
- **RQ3:** How effective is specification-level analysis in detecting and reducing API design flaws, inconsistencies, and compliance violations before deployment?

The originality of this work lies in its integrated approach combining Model-Driven Engineering (MDE), specification analysis, and automated compliance verification. By treating OpenAPI specifications as first-class models for reasoning about API behavior, constraints, and policies, this doctoral thesis supports the systematic detection, validation, and prevention of API design flaws, inconsistencies, and policy violations before deployment. Grounded in empirical observations of real-world APIs, the approach also enables continuous API quality assurance throughout the software lifecycle.

Methodology

To address **RQ1**, the PhD candidate will conduct an **empirical study of thousands of REST APIs** collected from platforms such as APIs.guru, RapidAPI, and GitHub in order to identify recurring design flaws, specification inconsistencies, and quality issues. These observations will support the **design of an enriched modeling framework** for REST APIs, based on OpenAPI, capable of explicitly capturing API constraints, quality properties, business rules, and regulatory policies. To address **RQ2**, the proposed modeling framework and associated tooling will be used to **generate enriched API specifications embedding explicit constraints related to data validation, regulatory policies, versioning rules, consistency, and constraints**. These enriched specifications aim to make API expectations and constraints explicit, thereby supporting automated reasoning, validation, and analysis of REST APIs. The proposed framework will also provide a foundation for advanced verification and testing mechanisms, including automated contract validation, specification consistency checking, policies compliance verification, regression testing and backward compatibility analysis. To address **RQ3**, the PhD candidate will integrate the verification and testing mechanisms developed in previous stages into **continuous integration and deployment (CI/CD) pipelines**, enabling continuous API quality assurance and automated compliance verification throughout the API lifecycle. This integration aims to detect and prevent API inconsistencies, specification drift, design regressions, misconfigurations, and policy violations in evolving REST APIs.

The expected outcomes include a **formal modeling framework, automated detection and verification tools, enriched API specifications, and practical strategies for continuous API quality assurance**. Experimental validation on both open-source and industrial APIs will assess the consistency, robustness, and effectiveness of these mechanisms in identifying and mitigating privacy risks.

Skills

- Master's degree in software engineering or a closely related field
- Knowledge in Web Development, OpenAPI and/or Model-Driven Engineering (MDE) and/or Web Privacy
- Interest in software quality, API engineering, software architecture, or automated verification
- Good programming skills
- Ability to work autonomously
- Willingness to learn about new technologies/techniques
- Good writing and communication skills
- Ability to propose, present, and discuss new ideas

The PhD candidate will be encouraged to publish in top-tier conferences such as ICSE, MODELS, ASE WebConf, and WWW conference, and to release the developed tools as open-source software (GPL or Apache), facilitating technology transfer to industry. Experiments on real-world cases will demonstrate the effectiveness of the proposed tools.

As is a common practice in the DiverSE research team, all source code will be open-sourced using either the GPL or Apache Licenses. It is expected that the doctoral student participate in related open-source communities. This should also assist in the technological transfer from academic prototypes to industrial-ready

tools. Experimentations to demonstrate the effectiveness of developed tools on real-world issues are actively encouraged and expected.

References

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2. H. Cao et al., "Automated Generation of REST API Specification from Plain HTML Documentation," ICSOC, 2017.
3. H. Ed-douibi et al., "Example-Driven Web API Specification Discovery," ECMFA, Springer, 2017.
4. S. M. Sohan et al., "SpyREST: Automated RESTful API Documentation Using an HTTP Proxy Server (N)," IEEE/ACM ASE, 2015.
5. S. Challita et al., "A Precise Model for Google Cloud Platform," IEEE IC2E, 2018.

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Mots-clés : REST APIs, OpenAPI, Model-Driven Engineering